

Trainings ▾

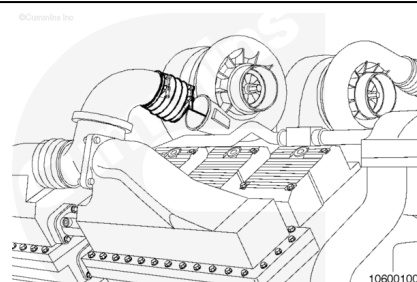
Remove

QSK45 Single Stage

Loosen the hose clamps on the hose between the air crossover and turbocharger.

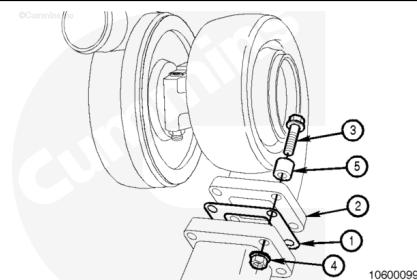
Slide the hose onto the air crossover.

Note : Do not disconnect the air crossover from the aftercooler.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note the position of the spacers and direction of the capscrews reassembly.

Remove the four capscrews (3), spacers (5), and flange nuts (4).

Remove the turbocharger (2) and gasket (1).

Discard the gasket (1), capscrews (3), and flange nuts (4).

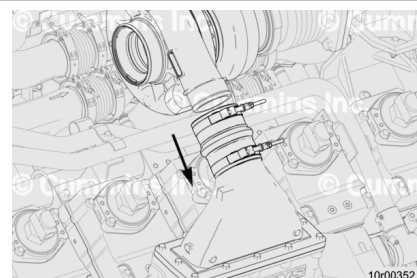
QSK60 Single Stage Industrial and Power Generation

Note : The QSK60 Single Stage has each high-pressure turbocharger supplying its own aftercooler cover.

Loosen the hose clamps on the hose between the air crossover and turbocharger.

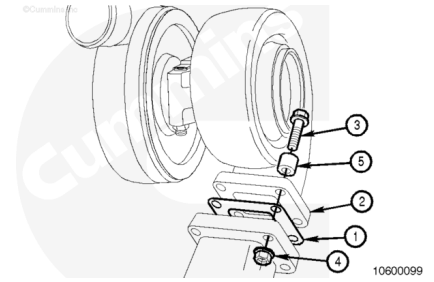
Slide the hose onto the aftercooler inlet connection.

Note : Do not disconnect the air crossover from the aftercooler.



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This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note the position of the spacers and direction of the capscrews reassembly.

Remove the four capscrews (3), spacers (5), and flange nuts (4).

Remove the turbocharger (2) and gasket (1).

Discard the gasket (1), capscrews (3), and flange nuts (4).

Cover openings. Use protective caps from Air Handling Clean Care Kit, Part Number 5394292.

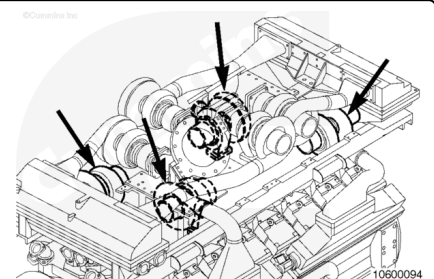
QSK60 Two Stage Industrial and Power Generation

High-pressure turbochargers are used on all QSK45 and QSK60 engines and are mounted directly to the exhaust manifold. The QSK60 two-stage engine also has four low-pressure turbochargers.

Removal of a high-pressure turbocharger on a QSK60 two-stage engine often requires the removal of low-pressure turbochargers and/or other components.

Determine which of the four high-pressure turbochargers to remove.

- Left bank rear
- Right bank front
- Left bank front
- Right bank rear.



The left bank rear and right bank front turbochargers are located under the low-pressure turbochargers. Remove the low-pressure turbocharger directly above the high-pressure turbocharger that requires removal. Refer to Procedure 010-035 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html>)

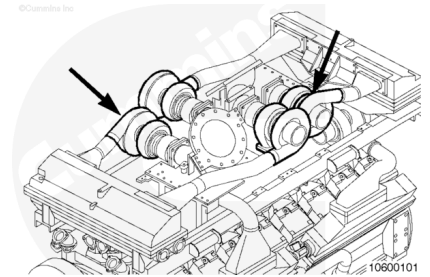
The left bank front and right bank rear turbochargers are located under the exhaust outlet connection and require removal of the exhaust outlet and mounting base, as detailed in this procedure.

Left Bank Rear or Right Bank Front

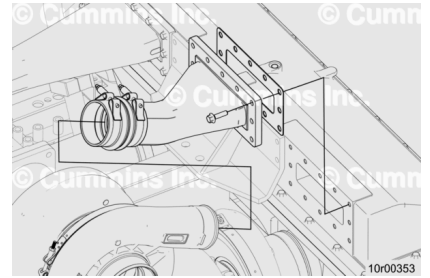
The left bank rear and right bank front turbochargers are located under the low-pressure turbochargers.

Remove the low-pressure turbocharger directly above the high-pressure turbocharger requiring removal. Refer to Procedure 010-035 in Section 10.

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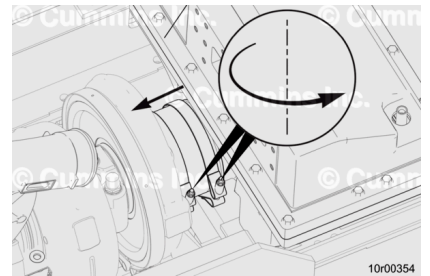


Remove the intercooler intake air crossover, located between the low-pressure turbocharger and the intercooler, to the intercooler.

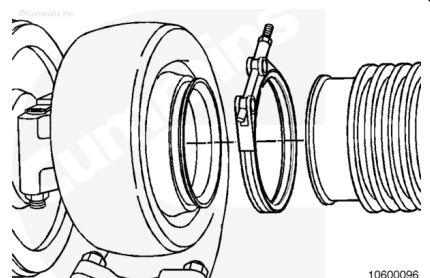


Loosen the two hose clamps on the hose between the turbocharger inlet and the intercooler.

Slide the hose toward the turbocharger.



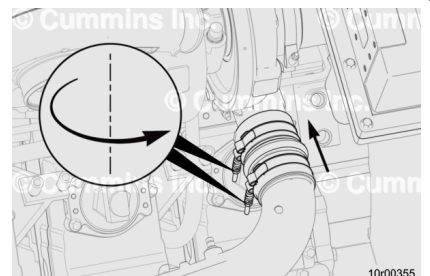
Remove the V-band clamp at the bellows to the turbine side of the turbocharger.



Loosen the hose clamps on the hose between the air crossover and turbocharger.

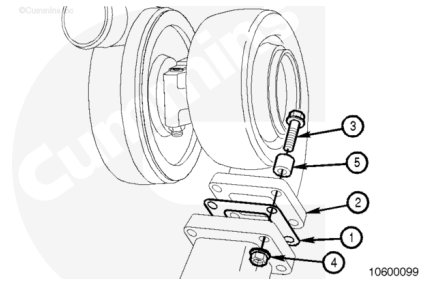
Slide the hose down the turbocharger outlet.

Note : Do not disconnect the air crossover from the aftercooler.



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Note the position of the spacers and direction of the capscrews reassembly.

Remove the four capscrews (3), spacers (5), and flange nuts (4).

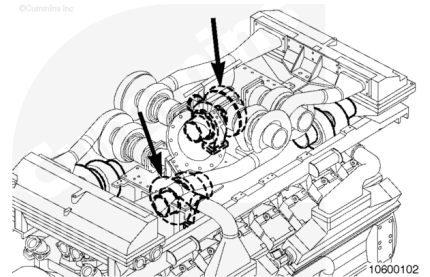
Remove the turbocharger (2) and gasket (1).

Discard the gasket (1), capscrews (3), and flange nuts (4).

Left Bank Front or Right Bank Rear

The left bank front and right bank rear turbochargers are located under the exhaust outlet connection and require removal of the exhaust outlet and mounting base.

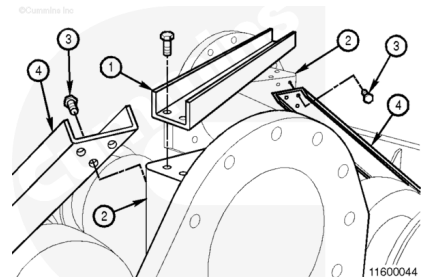
Remove the exhaust outlet and mounting base.



Remove the four capscrews from each end of the top cross support brace (1) between the two exhaust outlet connections (2).

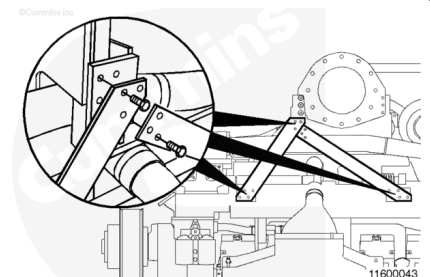
Remove the top cross support brace (1).

Remove the three capscrews (3) from each end of the two inboard support struts (4).



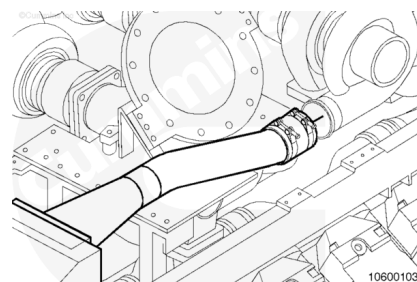
Remove the seven capscrews from each of the support braces between the exhaust outlet mounting base and the two-stage frame.

Remove the support braces.



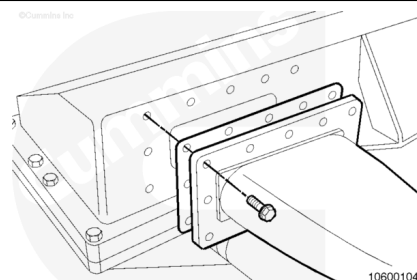
Loosen the clamps on the hose between the low-pressure turbocharger and the intercooler intake crossover.

Slide the hose and clamps onto the air crossover.

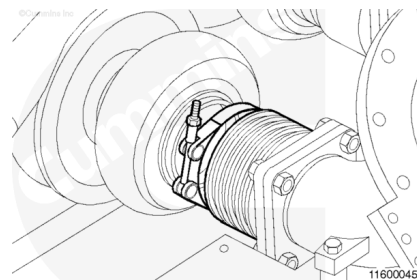


Remove the twelve capscrews connecting the intercooler intake crossover to the intercooler.

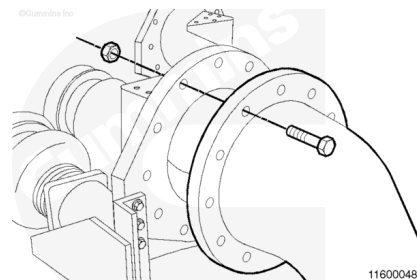
Remove and discard the gasket.



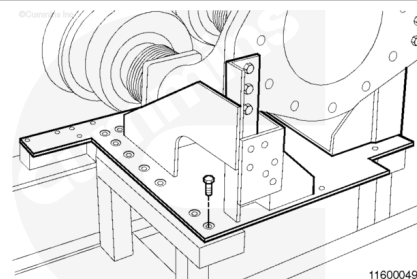
Disconnect, at the turbocharger, the V-band clamps connecting the bellows on the legs of the exhaust outlet flange to the low-pressure turbochargers.



Disconnect the exhaust piping from the exhaust outlet flange.

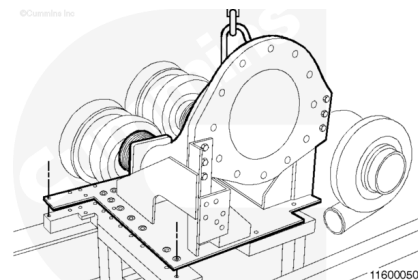


Remove the thirty capscrews connecting the exhaust outlet flange mounting base to the two-stage frame.



⚠ WARNING ⚠

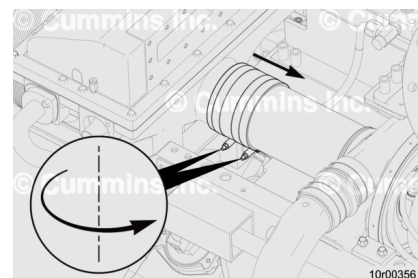
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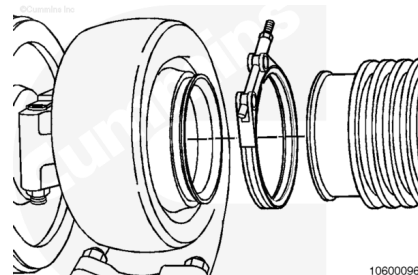
Remove the exhaust outlet flange and mounting base from the two-stage frame.

Loosen the two hose clamps on the hose between the turbocharger inlet and the intercooler.

Slide the hose toward the turbocharger.



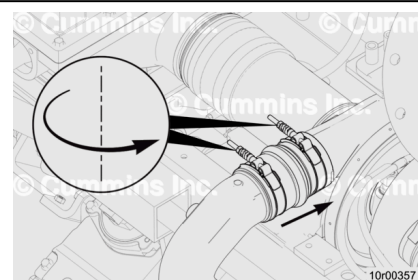
Remove the V-band clamp at the bellows to the turbine side of the turbocharger.



Loosen the hose clamps on the hose between the air crossover and turbocharger.

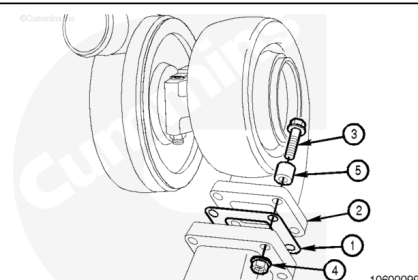
Slide the hose down the turbocharger outlet.

Note : Do not disconnect the air crossover from the aftercooler.



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Note the position of the spacers and direction of the capscrews reassembly.

Remove the four capscrews (3), spacers (5), and flange nuts (4).

Remove the turbocharger (2) and gasket (1).

Discard the gasket (1), capscrews (3), and flange nuts (4).

Last Modified: 18-Oct-2022
